Progressive Pseudo-label Self-balancing Towards Unsupervised Vision-Language Models Adaptation

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Abstract

Enhancing Vision-Language Models (VLMs) on downstream tasks with unlabeled data has attracted increasing attention. Constructing a pseudo-labeled dataset by pseudo-labeling is a common method. Nonetheless, owing to the inherent class-wise prediction bias in VLMs, they tend to generate long-tailed pseudo-labels that are inconsistent with the true label distribution. Motivated by this, we first divide all classes into pseudo-head, pseudo-middle, and pseudo-tail classes based on the pseudo-label quantity. Accordingly, we propose a **P**rogressive **P**seudo-label **S**elf-**B**alancing (PPSB) framework, which measures the class-wise prediction bias and calculates quantity upper bounds for certain classes to construct relatively balanced pseudo-labeled dataset, while introducing fewer incorrect pseudo-tail labels. In this way, the class-wise prediction bias is progressively corrected and the pseudo-label dataset scale increases autonomously during training. Furthermore, the confidence scores generated by VLMs are not entirely reliable. Therefore, we design a neighborhood consistency filtering mechanism and a visual classifier to improve the pseudo-label accuracy. We theoretically prove that our method can significantly reduce the generalization error under certain conditions. Extensive experiments across six benchmarks under three learning paradigms demonstrate that our method outperforms state-of-the-art methods by average **3.84**% in accuracy. The code is available at https://github.com/siqin-seu/PPSB-NeurIPS-26.

1 Introduction

Vision-Language Models (VLMs) [1, 2, 3, 4] are pre-trained on large-scale image-text pairs, which align images with corresponding textual descriptions and bridge the gap between visual and textual modalities. Recent studies [5, 6] have shown that VLMs still require labeled data to improve performance on downstream tasks. However, obtaining substantial labeled data is impractical and expensive [7, 8], so how to fine-tune VLMs with unlabeled data has attracted increasing attention.

Recent studies [9, 10] rely on the zero-shot capability of VLMs to generate pseudo-labels for unlabeled data. Nonetheless, VLMs tend to generate long-tailed pseudo-labels that are inconsistent with the true label distribution (See Fig.1). The bias mainly arises from two factors: (i) During pre-training, certain classes appear more frequently in the training data, leading VLMs to exhibit a stronger prediction bias towards these classes [11]. (ii) VLMs exploit class names to build a textual classifier for zero-shot classification, but some class names fail to accurately represent the underlying semantics of corresponding classes, leading to their under-prediction [10]. For instance, in the FGVC-Aircraft dataset, certain class names (e.g., "747-300") do not precisely convey the semantic meaning of the corresponding class ("Boeing 747-300"). To model class-wise imbalance in

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pseudo-labels, we divide all classes into pseudo-head, pseudo-middle, and pseudo-tail classes (Fig. 1). The prefix "pseudo-" is adopted because these classes are divided by pseudo-label quantity.

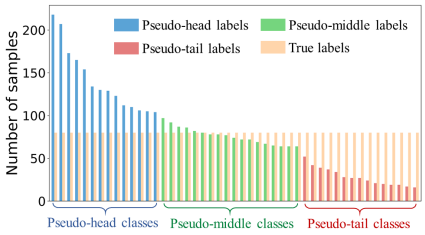


Figure 1: Long-tailed pseudo-label distribution on the dataset DTD.

Fine-tuning VLMs on imbalanced pseudo-labels inevitably leads to inferior performance [7]. Existing methods address this issue by constructing balanced pseudo-labeled dataset. Typically, FPL [9], and GRIP [9] filter the top-k confident pseudo-labeled samples for each class to construct pseudo-labeled dataset. CPL [7], AiR [10], and CAP [8] are built on the basis of GRIP or FPL. Among these methods, CPL generates partial labels [12] instead of hard pseudo-labels. CAP additionally incorporates a subset of high-confidence samples from the remaining unlabeled data and alleviates bias via confusion-aware calibrated margin. AiR exploits diffusion-generated images to build an auxiliary classifier and mitigate the bias of VLMs. However, this balanced strategy aligns the pseudo-label quantity of all classes, resulting in more incorrect pseudo-tail labels (See Fig. 2) and consequently impeding model training. Furthermore, existing methods merely rely on unreliable confidence scores [11] to filter pseudo-labels, not fully exploiting the representation information provided by samples.

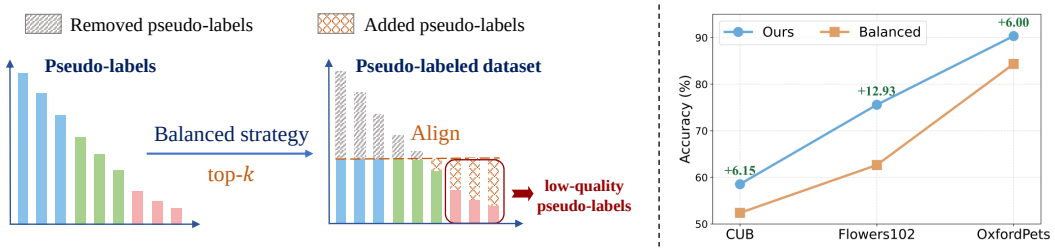


Figure 2: *left:* Balanced strategy aligns the pseudo-label quantity of all classes, resulting in more incorrect pseudo-tail labels. *right:* Pseudo-tail label accuracy across three datasets. Existing balanced strategy reduces the accuracy.

To address these challenges, we propose a **P**rogressive **P**seudo-label **S**elf-**B**alancing (PPSB) framework. As illustrated in Fig. 3, PPSB avoids generating extra pseudo-tail labels. Instead, it leverages the division of long-tailed pseudo-labels to measure the class-wise prediction bias and calculate an upper bound $\mathcal{U}$ to control the pseudo-head and pseudo-middle label quantity, thereby constructing a relatively balanced pseudo-labeled dataset. In the early stage of training, the model suffers from severe bias, resulting in a small $\mathcal{U}$ and constructing a small-scale pseudo-labeled dataset. As training on the relatively balanced dataset, upper bound $\mathcal{U}$ increases with the class-wise prediction bias gradually corrected, and all samples are progressively incorporated into the pseudo-labeled dataset.

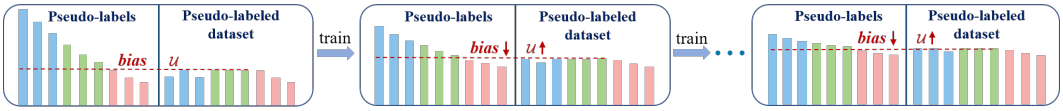


Figure 3: Overview of the progressive pseudo-label self-balancing framework.

To fully exploit the representation information of samples, we design the image-text neighborhood consistency filtering mechanism. Specially, it constructs an image-text representation matrix and captures its neighborhood information to filter high-quality pseudo-labels. Compared with the image representation matrix, our image-text representation matrix provides more discriminative neighborhood information (See Fig. 4). Furthermore, we introduce visual prototypes to build a visual classifier, and combine it with the class-wise prediction biased VLMs to generate more reliable pseudo-labels. Unlike AiR [10], we compute visual prototypes with reliable pseudo-labeled samples, without depending on generative models [13]. Finally, we theoretically prove that our method can significantly reduce the generalization error under certain conditions.

In summary, the main contributions of this work are as follows.

- We propose a **P**rogressive **P**seudo-label **S**elf-**B**alancing (PPSB) framework to avoid generating incorrect pseudo-tail labels and control the relative balance of the pseudo-labeled dataset. As training proceeds, PPSB gradually corrects the class-wise prediction bias and expands the pseudo-labeled dataset autonomously.
- We present an image-text neighborhood consistency filtering mechanism to select reliable pseudo-labels rather than merely rely on confidence scores. Assisted by filtered pseudo-labeled samples, a visual classifier is built to generate more reliable pseudo-labels.
- We provide a theoretical analysis of the generalization error for unsupervised VLMs adaptation, and prove that our method significantly reduces the generalization error under certain conditions.
- Comprehensive experiments on six commonly used benchmarks (DTD, FGVC-Aircraft, CUB, Flowers102, OxfordPets, and StanfordCars) under three learning paradigms validate that our method achieves new state-of-the-art performances with **3.84**% average improvement.

2 Related Work

2.1 Vision-Language Models

Vision-Language Models (VLMs) [1, 2, 3, 4] are pre-trained on large-scale image-text pairs with contrastive learning methods, which bridge the gap between visual and textual modalities. VLMs, such as CLIP [1], consist of an image encoder and a text encoder. A set of text prompts (e.g., "*a photo of [class]*") are encoded by the text encoder to form textual classifier for zero-shot classification. However, their performance degrades when facing domains that are rarely covered in the pre-training datasets. In recent years, numerous works have further improved the performance of VLMs on downstream tasks by fine-tuning them with annotated data [14, 15, 16]. In this paper, we mainly focus on fine-tuning CLIP [1] with unlabeled data to reduce the dependence on labeled data.

2.2 Prompt Tuning

Prompt tuning has emerged as a parameter-efficient alternative to full fine-tuning for adapting large-scale pre-trained models, such as CLIP [1] and GPT [17], to specific downstream tasks. In the context of VLMs, prompt tuning has been explored across different modalities. Textual prompt tuning, pioneered by CoOp [14], replaces manual templates with learnable context vectors in the textual branch to automate prompt engineering. Conversely, Visual Prompt Tuning [18] introduces a small budget of trainable parameters directly into the input sequence of the vision transformer's layers, providing an efficient adaptation strategy for the image encoder. In this paper, we fine-tune the prompts for class names in a manner similar to CoOp [14], enabling the CLIP model to adapt to downstream classification tasks.

2.3 Learning from pseudo-labels

In certain real-world downstream tasks, obtaining even a small portion of labeled data is expensive [7, 8, 10]. Recently, numerous studies have investigated fine-tuning downstream tasks with unlabeled data. These approaches mainly rely on the zero-shot capability of VLMs to generate pseudo-labels for unlabeled data. Existing pseudo-labeling methods construct balanced pseudo-labeled dataset to avoid the influence of imbalanced pseudo-labels generated by VLMs. FPL [9] select the top-k confident samples for each class. GRIP [9] regenerates pseudo-labels and filters top-k confident samples for classes within each iteration, where k increases with the predefined iterations. CPL [7] generates partial labels instead of hard pseudo-labels. CAP [8] additionally incorporates a subset of high-confidence samples from the remaining unlabeled data and alleviates bias via confusion-aware calibrated margin. AiR [10] exploits diffusion-generated images to build an auxiliary classifier. However, CPL, CAP, and AiR are still built on the basis of GRIP. This balanced strategy would introduce numerous incorrect pseudo-tail labels and hinder model training.

3 Methodology

Problem Definition. This paper addresses the problem of adapting VLMs to downstream tasks with only unlabeled data. Formally, given a set of unlabeled data $\mathcal{D}_{\mathrm{U}} = \{(x_i)\}_{i=1}^{N}$ containing N samples,

we focus on generating pseudo-label p_i for i-th sample within a class name space $\mathcal{Y} = \{c_j\}_{j=1}^{C}$ containing C classes. Then, we filter a portion of the pseudo-labels to construct the pseudo-labeled dataset $\mathcal{D}_L = \{(x_t, p_t)\}_{t=1}^{M}$ of size M, which is leveraged to fine-tune VLMs.

3.1 Progressive Pseudo-label Self-balancing Framework

Long-tailed pseudo-label distribution. The pseudo-labels generated by VLMs exhibit a typical long-tailed distribution (Fig. 1). Accordingly, we divide all classes into pseudo-head, pseudo-middle, and pseudo-tail classes based on the pseudo-label quantity of each class. The specific division method is as follows:

$$\begin{aligned} \mathcal{Y}_h &= \left\{ c_j \mid \tau_h \frac{N}{C} < N_{c_j} \right\}, \\ \mathcal{Y}_m &= \left\{ c_j \mid \tau_t \frac{N}{C} < N_{c_j} \leqslant \tau_h \frac{N}{C} \right\}, \\ \mathcal{Y}_t &= \left\{ c_j \mid N_{c_j} \leqslant \tau_t \frac{N}{C} \right\}, \end{aligned} \tag{1}$$

where N_{c_j} denotes the number of pseudo-labels for class c_j. $\mathcal{Y}_h$, $\mathcal{Y}_m$ and $\mathcal{Y}_t$ represent the class sets of pseudo-head, pseudo-middle and pseudo-tail classes, respectively. τ_h and τ_t are fixed thresholds for dividing $\mathcal{Y}_h$, $\mathcal{Y}_m$ and $\mathcal{Y}_t$. Sensitivity analysis of the two thresholds is shown in the Appendix F.1.

Pseudo-label self-balancing. If the model exhibits mild bias, it tends to predict a larger number of pseudo-middle and pseudo-tail labels. Therefore, the pseudo-middle and pseudo-tail label quantity is utilized to measure the class-wise prediction bias degree. We consider setting an upper bound $\mathcal{U}$ on the number of pseudo-head and pseudo-middle labels to control the relative balance of the pseudo-labeled dataset, which can be defined as:

$$\mathcal{U} = \left\lceil \frac{\sum_{c \in \mathcal{Y}_m \cup \mathcal{Y}_t} N_c}{|\mathcal{Y}_m \cup \mathcal{Y}_t|} \right\rceil. \tag{2}$$

$\mathcal{U}$ is the average number of pseudo-middle and pseudo-tail labels. A smaller $\mathcal{U}$ indicates more severe class-wise prediction bias.

For all pseudo-head labels, we sort them in descending order of confidence scores, and retain maximum $\mathcal{U}$ pseudo-labels for each class via image-text neighborhood consistency filtering mechanism (See ξ 3.2). For each pseudo-middle class, we observe that the precision of its pseudo-labels is relatively high (See Appendix F.2). Therefore, we directly retain up to $\mathcal{U}$ pseudo-labels in descending order of confidence scores. Since the number of pseudo-tail labels is small, we add all of them to the pseudo-labeled dataset.

Progressive process. *How does $\mathcal{U}$ control the pseudo-labeled dataset scale?* At the initial training stage, the small pseudo-tail label quantity results in a small $\mathcal{U}$, thus constructing a small scale and relative balanced the pseudo-labeled dataset. As the model is trained on the relative balanced labeled dataset, it gradually corrects the bias, and the number of pseudo-tail labels increases steadily. Meanwhile, $\mathcal{U}$ gradually increases as the model trains, and the pseudo-labeled dataset also expands continuously. Overall, we require no hyperparameters like iteration times to adjust the scale of the pseudo-labeled dataset. In contrast, the pseudo-labeled dataset expands autonomously based on the class-wise prediction bias degree.

3.2 Image-text Neighborhood Consistency Filtering Mechanism

Image-text representation matrix construction. Existing pseudo-labeling methods rely on confidence scores to filter high-quality pseudo-labels. However, the pseudo-label confidence scores generated by VLMs are not completely reliable. This raises the question:

How to filter reliable pseudo-labels beyond merely relying on confidence scores?

In this section, we attempt to select more reliable pseudo-labels exploiting image-text neighborhood information. Denote the image encoder and text encoder of VLMs as $\mathcal{I}(\cdot)$ and $\mathcal{T}(\cdot)$, respectively. We extract image feature vector $\mathbf{x}_i \in \mathbb{R}^d$ of each $x_i \in \mathcal{D}_U$ through $\mathcal{I}(\cdot)$ to form the image representation matrix $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_N]^\top \in \mathbb{R}^{N \times d}$ and extract text feature vector $\mathbf{c}_j \in \mathbb{R}^d$ of

each $c_j \in \mathcal{Y}$ through $\mathcal{T}(\cdot)$ to form the text representation matrix $\mathbf{C} = [\mathbf{c}_1, \mathbf{c}_2, \cdots, \mathbf{c}_C]^\top \in \mathbb{R}^{C \times d}$. We try to construct $\mathbf{W} \in \mathbb{R}^{N \times C}$, an image-text representation matrix, to obtain more discriminative representation for each sample. The optimization problem for solving $\mathbf{W}$ is given as follows:

$$\min_{\mathbf{W}} \|\mathbf{X} - \mathbf{W}\mathbf{C}\|_F^2 + \gamma \|\mathbf{W}\|_F^2, \tag{3}$$

where γ is a regularization coefficient that prevents $\mathbf{W}$ from falling into a trivial solution. Problem 3 is a standard ridge regression problem, whose closed-form solution is given by:

$$\mathbf{W} = \mathbf{X}\mathbf{C}^\top \left(\mathbf{C}\mathbf{C}^\top + \gamma \mathbf{E}_C\right)^{-1}, \tag{4}$$

where $\mathbf{E}_C \in \mathbb{R}^{C \times C}$ is an identity matrix. Each row of $\mathbf{W}$ represents the image-text feature vector of sample x_i, and is essentially the set of coefficients for a linear combination of text feature vectors guided by $\mathbf{x}_i$.

Neighborhood consistency filtering. For sample x_i, we extract its representation $\mathbf{w}_i$ from $\mathbf{W}$, and calculate the cosine similarity $cos(\mathbf{w}_i, \mathbf{w}_j)$ between x_i and each sample x_j $(i \neq j)$ from $\mathcal{D}_U$. Next, search for k samples $(x_1, \cdots, x_k)$ with the maximum cosine similarity to x_i. If the following equality holds:

$$\sum_{t=1}^{k} \mathbb{I}[p_t = p_i] = k, \tag{5}$$

then the pseudo-label p_i of x_i is considered reliable and added to pseudo-labeled dataset. $\mathbb{I}[\cdot]$ is an indicator function that takes 1 if the equality in it holds, and 0 otherwise.

Why we exploit image-text representation matrix $\boldsymbol{W}$ instead of image representation matrix $\boldsymbol{X}$? There are two main reasons:

- VLMs are trained on image-text pairs, and their zero-shot classification capability mainly stem from the alignment between visual and textual modality. Therefore, the cross-modal representation matrix $\mathbf{W}$ can better capture the differences among different samples. As shown in the Fig. 4, $\mathbf{W}$ achieves higher clustering accuracy than $\mathbf{X}$.
- During model training, we adopt a prompt tuning method similar to CoOp to fine-tune the prompts for class names, thereby yielding the $\mathbf{C}$ with more accurate semantic representations. Subsequently, we can solve $\mathbf{W}$ with higher clustering accuracy (See Fig. 4) via Eq. 4 during training.

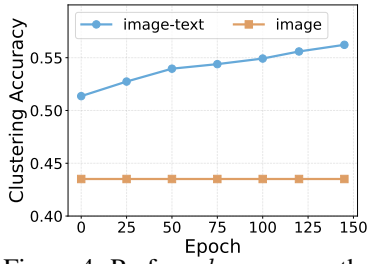


Figure 4: Perform k-means on the image representation matrix $\mathbf{X}$ and the image-text representation matrix $\mathbf{W}$ of the DTD dataset. $\mathbf{W}$ achieves higher accuracy.

3.3 Pseudo-labels Regeneration Assisted by Visual Prototypes

Original VLMs compute the class probability distribution $\hat{y}_i$ and pseudo-label p_i of sample $x_i \in \mathcal{D}_U$ with the textual classifier $\mathbf{C}$ as follows:

$$\hat{y}_i = \sigma(\mathbf{C}\mathbf{x}_i), \quad p_i = \arg\max_{c_j \in \mathcal{Y}} \left(\hat{y}_i^{c_j}\right), \tag{6}$$

where $\sigma(\cdot)$ denotes the softmax function and $\hat{y}_i^{c_j}$ denotes the probability of class c_j. For each class $c_j \in \mathcal{Y}$, we compute its visual prototype $\mathbf{I}_{c_j} \in \mathbb{R}^d$ with reliable pseudo-labeled samples in $\mathcal{D}_L$:

$$\mathbf{I}_{c_j} = \frac{1}{N_{c_j}} \sum_{t=1}^{M} \mathbb{I}[p_t = c_j] \mathbf{x}_t, \tag{7}$$

where $N_{c_j} = \sum_{t=1}^{M} \mathbb{I}[p_t = c_j]$. Then, a visual classifier $\mathbf{I} = [\mathbf{I}_1, \mathbf{I}_2, \cdots, \mathbf{I}_C]^\top \in \mathbb{R}^{C \times d}$ is built by concatenating all visual prototypes together. With the aid of visual classifier $\mathbf{I}$, we compute the class probability distribution of sample $x_i \in \mathcal{D}_U$:

$$\hat{y}_i = f(\mathbf{x}_i) = \sigma(\alpha \mathbf{C}\mathbf{x}_i + \mathbf{I}\mathbf{x}_i), \tag{8}$$

where α is a trade-off parameter. By fusing the logits from the visual and textual classifiers, we alleviate the bias of textual classifier with visual prototypes to regenerate more reliable pseudo-labels via Eq. 6. It should be noted that after regenerating pseudo-labels, we need to adjust the relative balance of the pseudo-labeled dataset.

3.4 Model Training and Prediction

Following CAP [8], we utilize LLMs to generate class descriptions to replace the original class names, thereby alleviating the concept mismatch problem. During model training, the image encoder and text encoder of VLMs are frozen, and only the prompts for the class descriptions are fine-tuned. We train the model with the cross-entropy loss function:

$$\mathcal{L}_{cro} = \mathbb{E}_{x_t \in \mathcal{D}_L} \left[-\sum_{c=1}^{C} \mathbb{I}\left[p_t = c\right] \log\left(y_t^c\right) \right]. \tag{9}$$

The detailed training process is shown in Appendix A. After model training, for a test sample x with image embedding $\mathbf{x}$, we calculate its logits via Eq. 8 and predict its class via Eq. 6.

3.5 Theoretical Analysis

We prove that our method can significantly reduce the generalization error under some conditions. Let $\ell_{avg} = \frac{1}{M_s}\left[\sum_{t=1}^{M_s} \ell_{cro}(f(x_t), p_t)\right]$ be the training loss on the M_s pseudo-labeled samples at the s-th training step, where p_t denotes the pseudo-label. Denote the pseudo-labeling error rate at the s-th training step as ϵ_s. Define the model empirical risk R at the s-th training step as:

$$R_s = R_{s-1} - G_s + I_{\epsilon_s} + I_{M_s}, \tag{10}$$

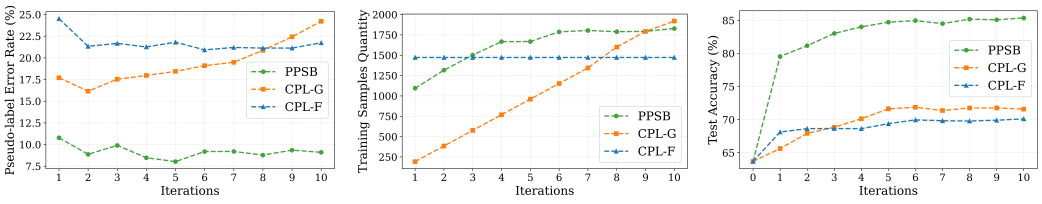
where $G_s \geq 0$ quantifies the empirical risk reduction achieved, and I_{ϵ_s} and I_{M_s} represent the impact of incorrect pseudo-labels and number of training samples, respectively. Let the function space $\mathcal{H}_c$ for the class $c \in \{1, \ldots, C\}$ be $\{h : x \longmapsto f_c(x) | f \in \mathcal{F}\}$, where $f_c(x)$ denotes the predicted probability of the c-th class and $\mathcal{F}$ denotes the function set of model f. Let $\mathcal{R}_{M_s}(\mathcal{H}_c)$ be the expected Rademacher complexity [19] of $\mathcal{H}_c$ with M_s training samples. The theorem is as follows:

Theorem 1 (Generalization Error)**.** *Suppose that the loss function $\ell_{cro}(f(x_t), c)$ is ρ-Lipschitz with respect to $f(x_t)$ for all $c \in \{1, \ldots, C\}$ and upper-bounded by B. Given the pseudo-labeling error rate $0 < \epsilon < 1$, for any $\delta > 0$, with probability at least $1 - \delta$, we have:*

$$R_S \leq R_0 - \sum_{s=1}^{S} G_s + B \sum_{s=1}^{S} \epsilon_s + 4\sqrt{2}\rho \sum_{s=1}^{S} \sum_{c=1}^{C} \mathcal{R}_{M_s}(\mathcal{H}_c) + 2B \sum_{s=1}^{S} \sqrt{\frac{\log \frac{2}{\delta}}{2M_s}}, \tag{11}$$

where S denotes the total number of training steps, R_S and R_0 represent the model empirical risk at the final training step and initial training step (with zero-shot pseudo-labels), respectively, and ϵ_s represents the pseudo-labeling error rate at the s-th training step.

The proof of Theorem 1 is provided in Appendix B. Based on Theorem 1, model empirical risk R_S mainly depends on three key factors, i.e., **the cumulative model empirical risk reduction $\sum_{s=1}^{S} G_s$, the cumulative pseudo-labeling error rate $\sum_{s=1}^{S} \epsilon_s$, and training samples quantity M_s at step s.** As shown in Figs. 5(a) and 5(b), our PPSB outperforms other methods, achieving both: (i) a consistently lower pseudo-labeling error rate ϵ_s (due to the combined contributions of the pseudo-label self-balancing framework, the neighborhood consistency filtering mechanism, and the visual classifier), and (ii) a growing number of training samples M_s gradually approaches the full dataset (due to the assistance of the progressive process). Moreover, training on a relatively balanced pseudo-labeled dataset with lower pseudo-label error rate yields higher cumulative model empirical risk reduction and ultimately reduces the generalization error significantly.



(a) Pseudo-labeling error rate. (b) Training samples quantity. (c) Test accuracy.

Figure 5: Comparison of pseudo-labeling error rate (a), training samples quantity (b), and test accuracy (c) among CPL-G [9], CPL-F [9], and PPSB on the dataset Flw. PPSB achieves lower pseudo-labeling error rate and comparable training samples quantity, obtaining higher test accuracy.

4 Experiments

To demonstrate the effectiveness of the proposed method, we conduct experiments under three different learning paradigms across six datasets and compare with eight SOTA methods. In addition, we conduct ablation studies and analytical experiments to explain why our method is effective.

4.1 Experiments Settings

Datasets and learning paradigms. We conducted experiments on six image classification datasets from distinct domains, including DTD (DTD.) [20], FGVC-Aircraft (FA.) [21], CUB (CUB.) [22], Flowers102 (Flw.) [23], OxfordPets (Pets.) [24], and StanfordCars (Cars.) [25]. To comprehensively evaluate our method, we consider three learning paradigms, including unsupervised learning (UL), semi-supervised learning (SSL) [26, 27, 28, 29], and transductive zero-shot learning (TRZSL) [30, 31]. More details about datasets and learning paradigms can be find in Appendix C.

Baselines. We compare our method with eight SOTA methods, namely, CLIP [1], FPL [9], GRIP [9], CPL-F [7], CPL-G [7], UEO [32], AiR [10], and CAP [8]. Appendix D shows more details about comparison methods. To ensure a fair comparison, we select the textual prompting tuning for each method except CAP which uses multi-modal prompts as MaPLe [33].

Implementation details. In our experiments, the image encoder of CLIP adopts ViT-B/32 and the prefix size is set to 16 for textual prompt learning. We set to fixed $\tau_h = 1.1$, $\tau_t = 0.9$, $\gamma = 1$, $k = 2$, and $\alpha = 1$. Qwen-plus is employed to enhance the descriptions of class names. We adopt SGD as the optimizer and conduct training for maximum 150 epochs. The learning rate is adjusted to 0.01 and decays following the cosine annealing rule. More implementation details can be find in Appendix E.

4.2 Main Results

Table 1 compares the performance of different methods under three learning paradigms. We report the average test accuracy over three random seeds 1, 2, 3. It can be observed that the proposed method consistently outperforms existing methods under the UL setting. Notably, our approach achieves significant improvements on Flowers102, FGVCAircraft, and CUB datasets, exceeding other baselines by **9.65**%, **4.09**%, and **4.26**%, respectively. This demonstrates that our method can effectively leverage unlabeled data to achieve superior performance. Under the SSL and TRZSL settings, PPSB outperforms other best method by average **2.62**% and **3.85**%, respectively. This suggests the capability of our method to leverage labeled data with unlabeled data, further validating its versatility regarding different paradigms. We also provide the results of different methods fine-tuned with VPT [18] in Appendix F.3.

Table 1: Comparison results of test accuracy (%) across six datasets under three learning paradigms. Values are expressed as mean_{std}. The highest accuracies are bold.

	Flowers102			FGVCAircraft			CUB			
Methods	SSL	UL	TRZSL	SSL	UL	TRZSL	SSL	UL	TRZSL	Avg.
CLIP [1]	$63.67_{0.00}$		$63.40_{0.00}$	$17.58_{0.00}$		$17.86_{0.00}$	$51.82_{0.00}$		$51.57_{0.00}$	$44.32_{0.00}$
FPL [9]	$75.57_{0.30}$	$68.22_{0.14}$	$86.25_{0.23}$	$21.03_{0.80}$	$17.90_{0.19}$	$28.15_{0.32}$	$55.83_{0.19}$	$53.43_{0.11}$	$63.12_{0.45}$	$52.17_{0.23}$
GRIP [9]	$88.42_{0.58}$	$67.43_{0.70}$	$85.49_{1.41}$	$22.95_{0.45}$	$16.78_{0.40}$	$30.08_{0.82}$	$57.52_{0.57}$	$51.50_{1.07}$	$65.07_{0.26}$	$53.92_{0.43}$
CPL-F [7]	$78.27_{0.03}$	$70.14_{0.32}$	$86.20_{0.45}$	$22.93_{0.48}$	$19.14_{0.13}$	$28.02_{0.31}$	$56.31_{0.24}$	$54.15_{0.13}$	$64.42_{0.21}$	$53.29_{0.23}$
CPL-G [7]	$90.48_{0.30}$	$72.14_{0.32}$	$89.65_{1.96}$	$23.38_{0.11}$	$17.88_{0.25}$	$30.74_{0.41}$	$57.54_{0.33}$	$53.60_{0.23}$	$66.09_{0.45}$	$55.72_{0.34}$
UEO [32]	$61.61_{1.75}$	$62.11_{1.06}$	$57.87_{3.50}$	$20.12_{0.26}$	$16.83_{1.16}$	$19.27_{0.69}$	$52.65_{0.30}$	$51.38_{0.84}$	$51.03_{1.06}$	$43.65_{0.36}$
CAP [8]	$87.72_{1.25}$	$76.78_{0.68}$	$87.49_{0.82}$	$21.32_{0.48}$	$17.26_{0.27}$	$27.52_{0.33}$	$57.73_{0.70}$	$55.54_{0.12}$	$60.68_{0.22}$	$54.67_{0.39}$
AiR [10]	$89.31_{0.11}$	$67.00_{0.63}$	$86.60_{1.65}$	$22.26_{0.23}$	$16.32_{0.20}$	$29.71_{0.36}$	$57.47_{0.16}$	$48.83_{0.54}$	$66.11_{0.13}$	$53.73_{0.30}$
PPSB (Ours)	$\mathbf{92.81}_{0.14}$	$\mathbf{86.43}_{0.81}$	$\mathbf{94.56}_{0.08}$	$\mathbf{25.45}_{0.63}$	$\mathbf{23.23}_{0.30}$	$\mathbf{34.21}_{0.57}$	$\mathbf{61.05}_{0.87}$	$\mathbf{59.80}_{0.29}$	$\mathbf{69.02}_{0.55}$	$\mathbf{60.73}_{0.48}$

	DTD			OxfordPets			StanfordCars			
Methods	SSL	UL	TRZSL	SSL	UL	TRZSL	SSL	UL	TRZSL	Avg.
CLIP [1]	$43.24_{0.00}$		$43.45_{0.00}$	$84.43_{0.00}$		$83.10_{0.00}$	$58.79_{0.00}$		$58.17_{0.00}$	$61.86_{0.00}$
FPL [9]	$52.09_{0.33}$	$47.00_{0.42}$	$57.27_{0.33}$	$87.92_{0.52}$	$88.52_{0.62}$	$87.83_{1.02}$	$64.20_{0.39}$	$60.32_{0.30}$	$70.13_{0.18}$	$68.36_{0.34}$
GRIP [9]	$59.71_{0.84}$	$47.96_{0.89}$	$66.88_{0.40}$	$87.18_{0.33}$	$87.70_{0.47}$	$89.91_{0.87}$	$64.47_{0.89}$	$57.33_{0.31}$	$73.36_{0.58}$	$70.50_{0.56}$
CPL-F [7]	$55.14_{0.73}$	$48.82_{0.67}$	$59.55_{0.80}$	$87.24_{0.73}$	$86.97_{0.12}$	$87.97_{1.32}$	$64.79_{0.02}$	$60.81_{0.21}$	$71.25_{0.20}$	$69.17_{0.49}$
CPL-G [7]	$60.90_{0.37}$	$50.99_{1.29}$	$67.62_{0.92}$	$85.93_{0.44}$	$83.56_{0.28}$	$83.17_{0.83}$	$65.78_{0.19}$	$60.33_{0.95}$	$74.15_{0.80}$	$70.27_{0.45}$
UEO [32]	$45.85_{0.41}$	$44.02_{0.81}$	$45.08_{1.32}$	$87.01_{1.22}$	$86.23_{1.25}$	$77.65_{1.63}$	$57.60_{0.60}$	$59.25_{0.12}$	$58.31_{0.83}$	$62.33_{0.49}$
CAP [8]	$61.40_{0.51}$	$56.04_{0.47}$	$69.92_{0.77}$	$\mathbf{91.37}_{0.19}$	$91.11_{0.20}$	$90.09_{0.19}$	$66.16_{0.25}$	$61.31_{0.36}$	$67.50_{0.50}$	$72.77_{0.42}$
AiR [10]	$58.98_{0.28}$	$48.35_{0.62}$	$66.39_{1.56}$	$87.95_{0.61}$	$88.46_{0.15}$	$89.46_{0.12}$	$64.37_{0.25}$	$56.85_{0.94}$	$73.21_{0.30}$	$70.45_{0.36}$
PPSB (Ours)	$\mathbf{63.36}_{0.70}$	$\mathbf{57.26}_{0.41}$	$\mathbf{71.24}_{0.42}$	$91.06_{0.31}$	$\mathbf{91.26}_{0.03}$	$\mathbf{90.98}_{0.30}$	$\mathbf{67.65}_{0.15}$	$\mathbf{62.23}_{0.36}$	$\mathbf{74.56}_{0.19}$	$\mathbf{74.40}_{0.29}$

4.3 Ablation Studies

Ablation of key components in our method. Our method consists of three components: the progressive pseudo-label self-balancing framework (PPSB-F), neighborhood consistency filtering mechanism (denoted as NCF), and visual classifier (denoted as VC). We conduct experiments on the DTD., Flw., and CUB. under the UL setting. As shown in Table 2, PPSB-F brings a significant gain of 11.25% on average across three datasets. Based on the baseline PPSB-F, incorporating the NCF and VC separately results in gains of 1.36% and 1.74%, respectively. When the two components are combined, our method achieves a remarkable performance improvement of 14.76% in accuracy, reinforcing the effectiveness and robustness of our design.

Table 2: Comparison results of test accuracy (%) w/ and w/o key components.

PPSB-F	NCF	VC	DTD.	CUB.	Flw.	Avg.
✗	✗	✗	44.99	43.49	70.51	52.99
✓	✗	✗	54.30	54.57	83.85	64.24
✓	✓	✗	56.43	56.31	84.07	65.60
✓	✗	✓	55.55	57.97	84.42	65.98
✓	✓	✓	**56.88**	**60.03**	**86.36**	**67.75**

Ablation of image encoder. To evaluate the generalizability of our method on different visual backbones, we conduct experiments with a larger image encoder ViT-14/L under the UL setting. As shown in Table 3, our method consistently outperforms other baselines by average **5.85**% in accuracy. The results illustrate that our method exhibits strong robustness and adaptability across various visual backbones, and can fully leverage the representation of diverse vision models.

Table 3: Comparison results of test accuracy (%) with image encoder ViT-14/L.

Method	Flw.	FA.	CUB.	DTD.	Pets.	Cars.
CLIP [1]	73.99	30.51	62.65	52.39	92.91	76.44
FPL [9]	79.67	31.17	67.19	55.95	94.71	79.82
GRIP [9]	76.46	28.86	69.24	55.58	95.53	79.08
CPL-F [7]	80.64	31.95	68.22	58.56	94.30	80.36
CPL-G [7]	80.20	32.43	71.52	57.55	95.31	81.02
UEO [32]	76.82	34.20	61.40	57.02	94.79	76.35
CAP [8]	87.80	33.48	70.76	61.59	95.17	70.95
AiR [10]	76.77	28.29	65.41	53.98	95.06	78.70
PPSB (Ours)	**95.10**	**40.08**	**75.62**	**64.57**	**96.07**	**83.43**

Table 4: Comparison results of test accuracy (%) on the foundation model OpenCLIP.

Method	Flw.	FA.	CUB.	DTD.	Pets.	Cars.
CLIP [1]	63.58	17.55	51.72	43.13	84.43	58.79
FPL [9]	68.98	19.38	61.16	51.86	88.60	82.91
GRIP [9]	69.27	18.72	59.71	52.23	87.92	82.32
CPL-F [7]	71.88	19.41	62.58	53.19	89.17	83.91
CPL-G [7]	72.58	20.07	61.49	54.78	89.56	84.69
UEO [32]	66.54	21.45	56.90	51.22	88.90	82.02
CAP [8]	74.14	18.45	59.02	52.02	**91.11**	66.34
AiR [10]	70.15	17.70	59.21	51.64	88.38	82.19
PPSB (Ours)	**87.57**	**26.67**	**66.34**	**59.99**	89.97	**86.17**

Ablation of hyperparameters. The hyperparameters of our method are fixed across all the datasets and experiments. We conduct a sensitivity study to explore the impact of hyperparameters (e.g. α and k) on the datasets Pets. and FA. in Appendix F.1. It can be observed that our method is insensitive to these hyperparameters, demonstrating the robustness of our method.

4.4 More Analysis

Progressive pseudo-label self-balancing framework improves pseudo-tail labels. We compare the pseudo-tail label accuracy with several methods (including GRIP, CPL-G, and AiR) on the final iteration (set $\tau_t = 1.0$). Fig. 6 shows that PPSB improves pseudo-tail label accuracy across all the datasets, which demonstrates that it effectively avoids introducing incorrect pseudo-tail labels.

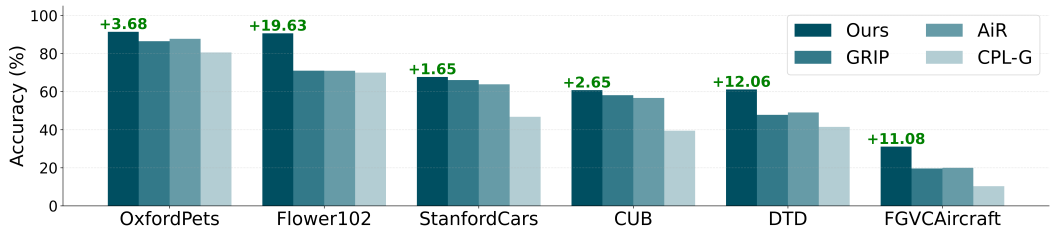
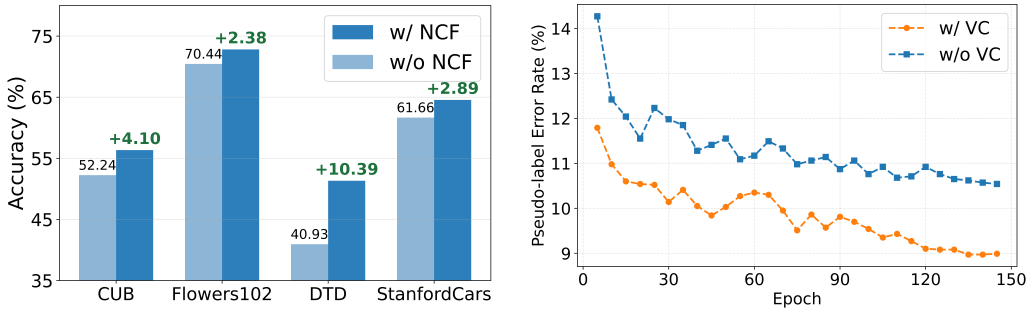


Figure 6: Comparison results of pseudo-tail label accuracy.

Neighborhood consistency filtering mechanism select more reliable pseudo-head labels. We compare pseudo-head label accuracy w/ and w/o neighborhood consistency filtering mechanism (NCF) across four datasets. As shown in Fig. 7(a), w/ NCF achieves a significant improvement of **10.39**% on the DTD and an average improvement of **4.94**% across four datasets.

Visual classifier consistently reduces the pseudo-label error rate. We conduct experiments on the dataset OxfordPets to compare the pseudo-label error rate w/ and w/o Visual Classifier (VC). As

shown in Fig. 7(b), w/ VC consistently exhibits lower pseudo-label error rate than w/o VC, which demonstrates that w/ VC can effectively generate more reliable pseudo-labels.



(a) Pseudo-head label accuracy w/ and w/o NCF.

(b) Pseudo-label error rate w/ and w/o VC.

Figure 7: (a) w/ NCF improve pseudo-head labels. (b) w/ VC reduce the pseudo-label error rate.

Evaluations on Imbalanced dataset. We explore the performance of different methods on the imbalanced datasets with varying imbalance ratios (10, 20). Experiments are conducted on four datasets under the UL setting. Existing methods perform quantity alignment for pseudo-labels of all classes. Instead, by controlling the relative balance of pseudo-labeled dataset, our method avoids introducing incorrect pseudo-tail labels. As shown in Table 5, PPSB consistently surpasses other baselines, which demonstrates the robustness of our method on imbalanced datasets.

Table 5: Comparison results of test accuracy (%) on the imbalanced datasets.

	Imbalance Ratio 10				Imbalance Ratio 20				
Methods	FA.	Pets.	Cars.	DTD.	FA.	Pets.	Cars.	DTD.	Avg.
FPL [9]	18.36	85.99	43.75	44.73	16.68	82.28	34.59	42.87	46.16
GRIP [9]	12.75	73.39	43.86	41.70	12.48	68.49	39.58	36.96	41.15
CPL-F [7]	17.22	86.64	58.36	47.55	16.74	85.71	56.77	44.89	51.74
CPL-G [7]	15.06	80.81	48.42	45.74	12.84	72.66	49.45	42.55	45.94
UEO [32]	14.64	81.65	57.40	39.41	14.67	80.43	57.07	37.28	47.83
CAP [8]	18.12	87.21	58.26	48.40	16.59	82.03	56.08	42.07	51.10
AiR [10]	15.09	74.32	44.06	39.20	11.94	64.67	38.36	38.56	40.78
PPSB (Ours)	**22.50**	**87.78**	**58.83**	**48.93**	**21.90**	**87.65**	**58.23**	**46.96**	**54.10**

Evaluations on different foundation models. To verify the effectiveness of our method on different foundation models under the UL setting, we conduct experiments on the OpenCLIP (LAION2B) with image encoder ViT-32/B across six datasets. Table 4 shows that PPSB exceeds other best baselines by average **5.59**%, which illustrates that our method can adapt to different foundation models.

Other experiments. We conduct more experiments (e.g. evolution of the pseudo-labels predictions, evolution of the pseudo-labeled dataset, statistical significance test, clustering performance of the image-text representation matrix, comparison of soft and hard pseudo-labels and evaluation with stronger vision-language models) in Appendix F, which further verifies the effectiveness of PPSB.

5 Conclusion

In this work, we delved into the issue of long-tailed pseudo-labels in unsupervised vision-language models adaptation. We proposed a progressive pseudo-label self-balancing framework that models long-tailed pseudo-label distribution and progressively corrects bias while introducing fewer pseudo-tail labels. By dynamically controlling the pseudo-head and pseudo-middle label quantity, PPSB constructs a relatively balanced pseudo-labeled dataset that expands autonomously during training. Furthermore, we introduced an image-text neighborhood consistency filtering mechanism and a visual classifier to improve pseudo-label accuracy. We also theoretically proved that our method reduces generalization error significantly under certain conditions. Extensive experiments across multiple benchmarks and learning paradigms demonstrate that our approach consistently outperforms existing methods by average **3.84**%, validating its effectiveness and robustness.

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A Training Process of PPSB

Overview. The overall workflow of our method PPSB can be divided into three steps:

(i) Leverage VLMs to generate pseudo-labels, and divide all classes into pseudo-head, pseudo-middle, and pseudo-tail classes. (ξ 3.1).

(ii) Calculate an upper bound for pseudo-head and pseudo-middle label quantity (ξ 3.1). Neighborhood consistency filtering mechanism is adopt to filter reliable pseudo-labels (ξ 3.2).

(iii) Update the visual classifier with filtered reliable pseudo-labeled samples, which assists in regenerating more reliable pseudo-labels, and re-execute Steps (i) and (ii) to control the relative balance of the pseudo-labeled dataset (ξ 3.3).

Steps (i-iii) are repeated until the model converge. Algorithm 1 shows the training process of PPSB.

Algorithm 1 Training process of PPSB

Input: Unlabeled datasets $\mathcal{D}_U$.
Output: Prompts $\mathbf{P}$ for class descriptions.
Initialize the parameters of $\mathbf{P}$ randomly.
for epoch $= 0, 1, \ldots$ **do**
  if epoch$\%5 = 0$ **then**
    Generate the pseudo-labels of $\mathcal{D}_U$.
    Step 1: Divide all classes into head, middle, and tail classes via Eq. 1.
      Calculate the upper bound $\mathcal{U}$ via Eq. 2.
    Step 2: Filter reliable pseudo-labels according to section 3.1.
    Update the visual classifier $\mathbf{I}$ via Eq. 7 and regenerate the pseudo-labels of $\mathcal{D}_U$.
    Repeat Steps 1 and 2 to obtain relatively balanced pseudo-labeled dataset $\mathcal{D}_L$.
  end if
  Calculate $\mathcal{L}_{cro}$ via Eq. 9.
  Update parameters $\mathbf{P}$ via gradient descent.
end for

B Proof of Theorem 1

Theorem 1. *Suppose that the loss function $\ell_{cro}(f(x_t), c)$ is ρ-Lipschitz with respect to $f(x_t)$ for all $c \in \{1, \ldots, C\}$ and upper-bounded by B. Given the pseudo-labeling error rate $0 < \epsilon < 1$, for any $\delta > 0$, with probability at least $1 - \delta$, we have:*

$$R_S \leq R_0 - \sum_{s=1}^{S} G_s + B \sum_{s=1}^{S} \epsilon_s + 4\sqrt{2}\rho \sum_{s=1}^{S} \sum_{c=1}^{C} \mathcal{R}_{M_s}(\mathcal{H}_c) + 2B \sum_{s=1}^{S} \sqrt{\frac{\log \frac{2}{\delta}}{2M_s}}, \tag{12}$$

where S denotes the total number of training steps, R_S and R_0 represent the model empirical risk at the final training step and initial training step (with pseudo-labels generated by zero-shot capability of VLMs), respectively, and ϵ_s represents the pseudo-labeling error rate at the s-th training step.

Proof. We first restate the definition of the model empirical risk R at the s-th training step, i.e.,

$$R_s = R_{s-1} - G_s + I_{\epsilon_s} + I_{M_s}, \tag{13}$$

where R_s denotes the model empirical risk at the s-th training step, $G_s \geq 0$ quantifies the empirical risk reduction achieved, and I_{ϵ_s} and I_{M_s} represent the impact of incorrect pseudo-labels and number of training samples, respectively. We derive I_{ϵ_s} and I_{M_s} below, omitting the subscript s for brevity.

Denote the true risk on the training dataset with respect to the classification model $f(x; \theta)$ as:

$$R(f) = \mathbb{E}_{(x,y)} \left[\ell_{cro}(f(x), c) \right]. \tag{14}$$

We aim to learn a good classification model by minimizing the empirical risk $\widehat{R}(f) = \frac{1}{M} \sum_{t=1}^{M} \ell_{cro}(f(x_t), c_t)$ on the labeled dataset $\mathcal{D}_L$.

Next, we derive the uniform deviation bound I_M between the true risk $R(f)$ and the empirical risk $\widehat{R}(f)$ using the following lemma.

Lemma 1. *Suppose that the loss function $\ell_{cro}(f(x), c)$ is ρ-Lipschitz with respect to $f(x)$ for all $c \in \{1, \ldots, C\}$ and upper-bounded by B. For any $\delta > 0$, with probability at least $1 - \delta$, we have:*

$$|R(f) - \widehat{R}(f)| \leq 2\sqrt{2}\rho \sum_{c=1}^{C} \mathcal{R}_M(\mathcal{H}_c) + B\sqrt{\frac{\log \frac{2}{\delta}}{2M}}, \tag{15}$$

where the function space $\mathcal{H}_c$ for the label $c \in \{1, \ldots, C\}$ is $\{h : x \longmapsto f_c(x) | f \in \mathcal{F}\}$.

Proof. In order to prove this lemma, we define the Rademacher complexity of the composition of loss function ℓ_{cro} and model $f \in \mathcal{F}$ with M training samples as follows:

$$\begin{aligned} &\mathcal{R}_M(\ell_{cro} \circ \mathcal{F}) \\ =&\mathbb{E}_{(x,c,\mu)} \left[\sup_{f \in \mathcal{F}} \sum_{t=1}^{M} \mu_t \Big(\ell_{cro}\big(f(x_t), c_t\big) \Big) \right] \\ \leq&\sqrt{2}\rho \sum_{c=1}^{C} \mathcal{R}_M(\mathcal{H}_c), \end{aligned} \tag{16}$$

where $\circ$ denotes the function composition operator, $\mathbb{E}_{(x,c,\mu)}$ denotes the expectation over x, c, and μ, μ denotes the Rademacher variable, $\sup_{f \in \mathcal{F}}$ denotes the supremum (or least upper bound) over the function set $\mathcal{F}$ of model f. The second line holds because of the Rademacher vector contraction inequality [34].

Then, we proceed with the proof by showing that the one direction $\sup_{f \in \mathcal{F}} R(f) - \widehat{R}(f)$ is bounded with probability at least $1 - \delta/2$, and the other direction $\sup_{f \in \mathcal{F}} \widehat{R}(f) - R(f)$ can be proved similarly. Note that replacing a sample (x_t, c_t) leads to a change of $\sup_{f \in \mathcal{F}} R(f) - \widehat{R}(f)$ at most $\frac{B}{M}$ due to the fact that ℓ_{cro} is bounded by B. According to the McDiarmid's inequality [19], for any $\delta > 0$, with probability at least $1 - \delta/2$, we have:

$$\sup_{f \in \mathcal{F}} R(f) - \widehat{R}(f) \leq \mathbb{E}\left[\sup_{f \in \mathcal{F}} R(f) - \widehat{R}(f) \right] + B\sqrt{\frac{\log \frac{2}{\delta}}{2M}}. \tag{17}$$

According to the result in [19] that shows $\mathbb{E}\left[\sup_{f \in \mathcal{F}} R(f) - \widehat{R}(f)\right] \leq 2\mathcal{R}_M(\mathcal{F})$, and further considering the other direction $\sup_{f \in \mathcal{F}} \widehat{R}(f) - R(f)$, with probability at least $1 - \delta$, we have:

$$\sup_{f \in \mathcal{F}} |R(f) - \widehat{R}(f)| \leq 2\sqrt{2}\rho \sum_{c=1}^{C} \mathcal{R}_M(\mathcal{H}_c) + B\sqrt{\frac{\log \frac{2}{\delta}}{2M}}, \tag{18}$$

which completes the proof.

□

In unsupervised learning, we cannot minimize the empirical risk $\widehat{R}(f)$ directly, since the true labels are inaccessible for $\mathcal{D}_L$. Therefore, we need to train the model with $\widehat{R}'(f) = \frac{1}{M}\sum_{t=1}^{M} \ell_{cro}(f(x_t), p_t) \geq \widehat{R}(f)$, where p_t denotes the pseudo-label of unlabeled sample x_t. Let $\hat{f} = \operatorname{argmin}_{f \in \mathcal{F}} \widehat{R}(f)$ be the empirical risk minimizer, and $f^* = \operatorname{argmin}_{f \in \mathcal{F}} R(f)$ be the true risk minimizer. Let $\ell_{all} = \frac{1}{M}\sum_{t=1}^{M} \ell_{cro}(f(x_t), p_t)$ be the training loss on M pseudo-labeled samples with pseudo-labeling error rate ϵ for our method.

Then, we can bound the difference I_{ϵ_s} between $\widehat{R}(f)$ and $\widehat{R}'(f)$ as follows.

Lemma 2. *Suppose that the loss function $\ell_{cro}(f(x), c)$ is ρ-Lipschitz with respect to $f(x)$ for all $c \in \{1, \ldots, C\}$ and upper-bounded by B. Given the pseudo-labeling error rate $0 < \epsilon < 1$, we have:*

$$0 \leq |\widehat{R}'(f) - \widehat{R}(f)| \leq B\epsilon. \tag{19}$$

Proof. Let's first expand $\widehat{R}'(f)$:

$$\begin{aligned}\widehat{R}'(f) &= \frac{1}{M}\sum_{t=1}^{M}\ell_{cro}(f(x_t),p_t)\\ &= \frac{1}{M}\sum_{t=1}^{M}\left[(1-\epsilon)\ell_{cro}(f(x_t),c_t)+\frac{\epsilon}{C-1}\sum_{c=1}^{C}\mathbb{I}(p_t\neq c_t)\ell_{cro}(f(x_t),p_t)\right].\end{aligned} \tag{20}$$

Next, we show its upper bound:

$$\begin{aligned}\widehat{R}'(f) &\leq \widehat{R}(f)+\frac{1}{M}\sum_{t=1}^{M}\left[\frac{\epsilon}{C-1}\sum_{c=1}^{C}\mathbb{I}(p_t\neq c_t)\ell_{cro}(f(x_t),c_t)\right]\\ &\leq \widehat{R}(f)+\frac{1}{M}\sum_{t=1}^{M}\left[\frac{\epsilon}{C-1}\sum_{c=1}^{C}\mathbb{I}(\hat{y}_j\neq y_j)B\right]\\ &\leq \widehat{R}(f)+B\epsilon,\end{aligned} \tag{21}$$

and when $\epsilon\to 0$, it reaches the lower bound $\widehat{R}(f)$, which concludes the proof.

□

Based on the above lemmas, for any $\delta>0$, with probability at least $1-\delta$, at the s-th training step, we have:

$$\begin{aligned}R_s(\hat{f}) &\leq \widehat{R}_s(\hat{f})+2\sqrt{2}\rho\sum_{c=1}^{C}\mathcal{R}_{M_s}(\mathcal{H}_c)+B\sqrt{\frac{\log\frac{2}{\delta}}{2M_s}}\\ &\leq \widehat{R}'_s(\hat{f})+2\sqrt{2}\rho\sum_{c=1}^{C}\mathcal{R}_{M_s}(\mathcal{H}_c)+B\sqrt{\frac{\log\frac{2}{\delta}}{2M_s}}\\ &\leq \widehat{R}'_s(f)+2\sqrt{2}\rho\sum_{c=1}^{C}\mathcal{R}_{M_s}(\mathcal{H}_c)+B\sqrt{\frac{\log\frac{2}{\delta}}{2M_s}}\\ &\leq \widehat{R}_s(f)+B\epsilon+2\sqrt{2}\rho\sum_{c=1}^{C}\mathcal{R}_{M_s}(\mathcal{H}_c)+B\sqrt{\frac{\log\frac{2}{\delta}}{2M_s}}\\ &\leq R_s(f)+B\epsilon+4\sqrt{2}\rho\sum_{c=1}^{C}\mathcal{R}_{M_s}(\mathcal{H}_c)+2B\sqrt{\frac{\log\frac{2}{\delta}}{2M_s}},\end{aligned} \tag{22}$$

where the first and last lines are based on Lemma 1, and the fourth lines are due to Lemma 2. The third line is by the definition of $\hat{f}$.

Then, we can derive $I_{\epsilon_s}+I_{M_s}$ as follows:

$$\begin{aligned}I_{\epsilon_s}+I_{M_s} &= |\widehat{R}'_s(f)-\widehat{R}_s(f)|+|R_s(f)-\widehat{R}_s(f)|\\ &= \widehat{R}'_s(f)-\widehat{R}_s(f)-R_s(f)+\widehat{R}_s(f)\\ &= \widehat{R}'_s(f)-R_s(f)\\ &\leq R_s(\hat{f})-R_s(f)\\ &\leq B\epsilon_t+4\sqrt{2}\rho\sum_{c=1}^{C}\mathcal{R}_{M_s}(\mathcal{H}_c)+2B\sqrt{\frac{\log\frac{2}{\delta}}{2M_s}},\end{aligned} \tag{23}$$

Substituting Eq. (23) into Eq. (13) yields:

$$\begin{cases} R_s \leq R_{s-1} - G_s + B\epsilon_s + 4\sqrt{2}\rho \sum_{c=1}^{C} \mathcal{R}_{M_s}(\mathcal{H}_c) + 2B\sqrt{\frac{\log\frac{2}{\delta}}{2M_s}}, \\ R_{s-1} \leq R_{s-2} - G_{s-1} + B\epsilon_{s-1} + 4\sqrt{2}\rho \sum_{c=1}^{C} \mathcal{R}_{M_{s-1}}(\mathcal{H}_c) + 2B\sqrt{\frac{\log\frac{2}{\delta}}{2M_{s-1}}}, \\ \quad \vdots \\ R_1 \leq R_0 - G_1 + B\epsilon_1 + 4\sqrt{2}\rho \sum_{c=1}^{C} \mathcal{R}_{M_1}(\mathcal{H}_c) + 2B\sqrt{\frac{\log\frac{2}{\delta}}{2M_1}}. \end{cases} \tag{24}$$

Finally, combining the sub-equations in Eq. (24) through summation yields:

$$R_S \leq R_0 - \sum_{s=1}^{S} G_s + B\sum_{s=1}^{S} \epsilon_s + 4\sqrt{2}\rho \sum_{s=1}^{S}\sum_{c=1}^{C} \mathcal{R}_{M_s}(\mathcal{H}_c) + 2B\sum_{s=1}^{S}\sqrt{\frac{\log\frac{2}{\delta}}{2M_s}}. \tag{25}$$

At this point, we have proven Theorem 1.

□

C Dataset and Learning Paradigm Details

C.1 Dataset Details

Experiments are conducted on six image classification datasets from distinct domains, including DTD [20], FGVC-Aircraft [21], CUB [22], Flowers102 [23], OxfordPets [24], and StanfordCars [25]. We present dataset statistics in Table 6. All datasets adopt the default training-test splits.

Table 6: Dataset statistics.

	DTD	FGVC-Aircraft	CUB	Flowers102	OxfordPets	StanfordCars
Class number	47	100	200	102	37	196
Training set size	3760	6667	5594	2040	3680	8144
Testing set size	1880	3333	5794	6149	3669	8041
Balanced	✓	✓	✗	✗	✗	✗

C.2 Learning Paradigm Details

In main results (ξ 4.2), we consider three learning paradigms, including unsupervised learning (UL), semi-supervised learning (SSL), and transductive zero-shot learning (TRZSL). We provide a detailed explanation of experimental settings for three learning paradigms, in line with [9].

- For Unsupervised Learning (UL), we only have access to unlabeled data. In this scenario, we initially rely on the zero-shot predictions of CLIP to obtain all pseudolabels without any manual annotation.
- For Semi-Supervised Learning (SSL), access to labeled data is limited. We assess the impact of pseudolabels in scenarios with a few labeled data and abundant unlabeled data, using two labeled samples per class.
- For Transductive Zero-Shot Learning (TRZSL), labeled data for certain target classes (seen classes) are provided in the downstream dataset. We set the ratio of seen to unseen classes at 62-38, with all pseudolabels generated from unseenclasses. It is noteworthy that in TRZSL, we report the harmonic mean of the accuracies of seen and unseen classes.

D Comparison Methods

We compare our method with eight SOTA methods, including CLIP [1], Few Pseudo-Labels (FPL [9]), Grow and Reffne Iteratively Pseudo-labels (GRIP [9]), Candidate Pseudo-Labeling with FPL (CPL-F [7]), Candidate Pseudo-Labeling with GRIP (CPL-G [7]), Universal Entropy Optimization (UEO [32]), Augmenting Discriminative Richness (AiR [10]), and Concept-Adaptive Pseudo-labeling (CAP [8]). The details of each method are presented as follows:

- **CLIP :** CLIP exhibits zero-shot capability for classification.
- **FPL :** FPL filters top-k confident pseudo-labels of each class to construct balanced pseudo-label dataset for model training.
- **GRIP :** GRIP pre-defines the iteration times. Within each iteration, it regenerates pseudo-labels and filters top-k confident pseudo-labels of each class to construct balanced pseudo-label dataset, where k increases linearly as iterations proceed.
- **CPL:** CPL uses intra- and inter-instance selected candidate pseudo-labels instead of hard pseudo-labels, effectively reducing label noise and class imbalance for better adaptation. **CPL-F** and **CPL-G** are implemented on the basis of FPL and GRIP, respectively.
- **UEO:** UEO is a universal entropy optimization method for realistic unsupervised CLIP fine-tuning. It uses sample confidence to minimize entropy for in-distribution samples and maximize marginal entropy for OOD samples.
- **AiR:** AiR utilizes LoRA-fine-tuned diffusion models to generate class-representative synthetic images and builds an auxiliary classifier fused with text classifier for high-quality pseudo-labels.
- **CAP:** CAP proposes concept alignment to fix semantic-visual misalignment, and a confusion-aware calibrated margin to improve discrimination between similar classes. It also adopts dual independent adapters to separately learn balanced pseudo-labels and high confidence unlabeled data.

E Implementation Details

We present implementation details in Table 7. Other methods follow the implementation in the original paper.

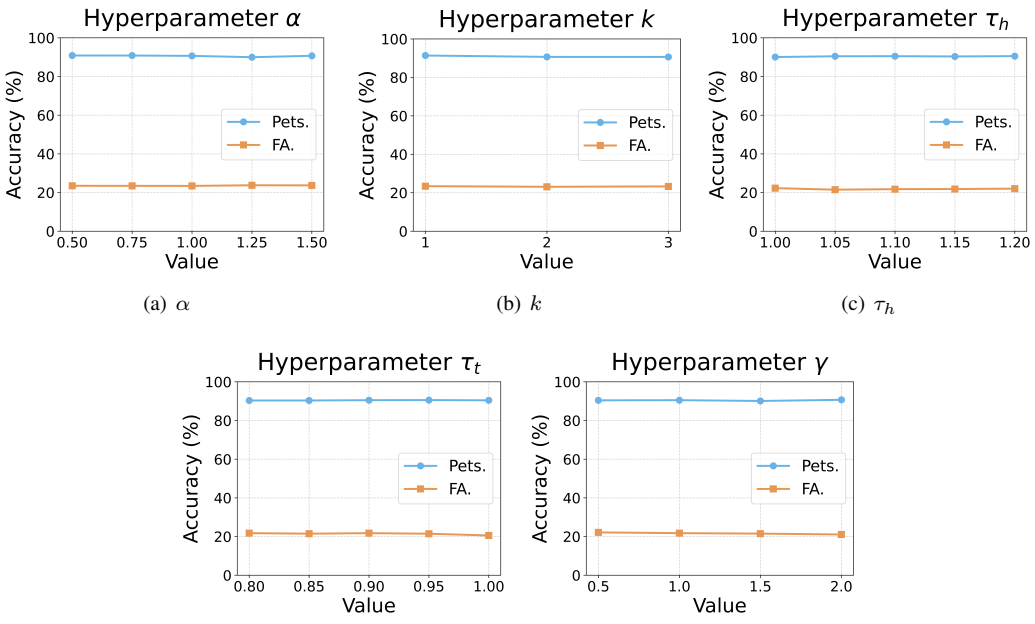
Table 7: Implementation Details

Training Setting		**Hyperparameters**	
Network	ViT-B / 32	Pseudo-head class threshold τ_h	1.1
Batch size	32	Pseudo-tail class threshold τ_t	0.9
Epoch	150	γ in Eq. 3	1
Optimizer	SGD	k in Eq. 5	2
Learning rate (LR)	0.01	α in Eq. 8	1
Warm epoch	2		
Warmup Learning rate	1e-4		
Weight decay	0.05		
LR scheduler	CosineAnnealingLR		
LLMs	Qwen-plus		

F More Experiment Results and Analysis

F.1 Sensitivity Study of the Hyperparameters

We conduct a sensitivity study to explore the impact of hyperparameters α, k, τ_h, τ_t, and γ on the datasets Pets. and FA. . Fig. 8 shows that our method is insensitive to these hyperparameters, further demonstrating the robustness of our method.



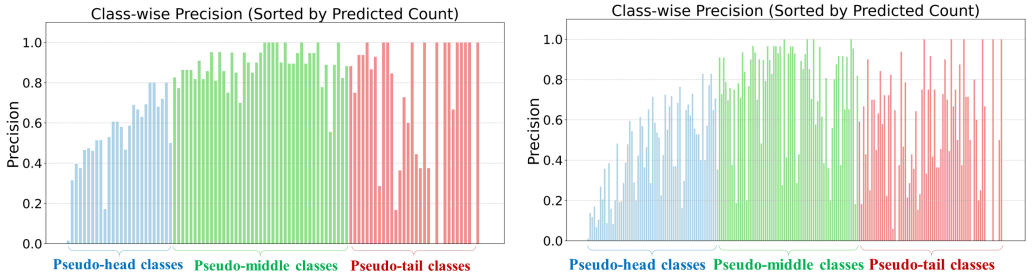
(a) α (b) k (c) τ_h

(d) τ_t (e) γ

Figure 8: Sensitivity analysis of α, k, τ_h, τ_t, and γ.

F.2 Precision of Pseudo-labels

As shown in Fig. 9, we observe that pseudo-middle classes achieve higher precision than pseudo-head classes. Therefore, the confidence scores of pseudo-middle labels are more reliable than pseudo-head labels and we filter them via confidence scores directly without neighborhood consistency filtering mechanism.



(a) Flowers102 (b) CUB

Figure 9: Precision of pseudo-labels across different classes which sorted by pseudo-label quantity.

F.3 Comparison of Methods Fine-tuned with VPT

We fine-tune all comparison methods with VPT [18] except CAP [8], due to its results with MaPLe [33] already shown in Table 1. As shown in Table 8, our method PPSB also achieves competitive performance on six datasets under three learning paradigms, exceeding SOTA baselines by average **3.70**% in accuracy. This demonstrates that PPSB can adapt to different fine-tuning methods.

Table 8: Test accuracy (%) of methods fine-tuned with VPT across six datasets under three learning paradigms. Values are expressed as mean ± standard deviation. The highest accuracies are bold.

Methods	Flowers102			FGVCAircraft			CUB			
	SSL	UL	TRZSL	SSL	UL	TRZSL	SSL	UL	TRZSL	Avg.
FPL [9]	$70.22_{0.26}$	$67.50_{0.23}$	$79.02_{0.54}$	$20.84_{0.32}$	$19.35_{0.27}$	$26.00_{0.35}$	$52.78_{0.19}$	$53.25_{0.22}$	$56.11_{0.15}$	$49.45_{0.28}$
GRIP [9]	$71.68_{0.09}$	$63.65_{0.28}$	$78.52_{0.75}$	$19.90_{0.22}$	$18.12_{0.34}$	$30.56_{0.29}$	$49.72_{0.49}$	$50.72_{0.30}$	$56.02_{0.14}$	$48.77_{0.32}$
CPL-F [7]	$70.16_{0.07}$	$69.20_{0.24}$	$78.85_{0.22}$	$21.43_{0.59}$	$19.88_{0.11}$	$26.87_{0.41}$	$53.49_{0.48}$	$53.72_{0.40}$	$56.29_{0.39}$	$49.99_{0.32}$
CPL-G [7]	$72.72_{0.45}$	$67.89_{1.48}$	$79.81_{0.43}$	$19.72_{0.24}$	$18.47_{0.71}$	$\mathbf{31.08}_{0.47}$	$49.64_{0.40}$	$51.55_{0.20}$	$57.00_{0.61}$	$49.76_{0.55}$
UEO [32]	$54.11_{2.19}$	$56.28_{2.31}$	$49.01_{4.13}$	$18.97_{0.16}$	$18.95_{0.47}$	$18.48_{0.26}$	$49.88_{0.77}$	$50.80_{0.21}$	$48.03_{0.84}$	$40.50_{1.26}$
AiR [10]	$72.02_{0.39}$	$63.96_{0.15}$	$78.25_{0.51}$	$20.70_{0.32}$	$18.00_{0.12}$	$29.62_{0.42}$	$48.83_{0.46}$	$49.73_{0.21}$	$56.27_{0.37}$	$48.60_{0.33}$
PPSB (Ours)	$\mathbf{79.30}_{1.91}$	$\mathbf{73.55}_{0.37}$	$\mathbf{82.44}_{0.05}$	$\mathbf{23.77}_{0.51}$	$\mathbf{22.61}_{0.26}$	$30.08_{0.10}$	$\mathbf{56.26}_{0.04}$	$\mathbf{57.55}_{0.16}$	$\mathbf{57.96}_{0.23}$	$\mathbf{53.72}_{0.40}$

Methods	DTD			OxfordPets			StanfordCars			
	SSL	UL	TRZSL	SSL	UL	TRZSL	SSL	UL	TRZSL	Avg.
FPL [9]	$51.63_{0.58}$	$47.62_{0.32}$	$55.06_{0.61}$	$89.14_{0.21}$	$89.12_{0.14}$	$89.38_{0.10}$	$59.54_{0.28}$	$59.92_{0.39}$	$63.66_{0.47}$	$67.23_{0.34}$
GRIP [9]	$57.73_{0.73}$	$48.91_{0.23}$	$64.12_{0.71}$	$89.65_{0.08}$	$89.94_{0.52}$	$90.53_{0.07}$	$54.96_{0.19}$	$54.67_{0.33}$	$63.77_{0.25}$	$68.25_{0.35}$
CPL-F [7]	$52.62_{0.69}$	$51.06_{0.37}$	$55.72_{0.35}$	$88.51_{0.35}$	$87.83_{0.01}$	$87.22_{0.12}$	$59.32_{0.21}$	$59.55_{0.06}$	$63.27_{0.26}$	$67.23_{0.27}$
CPL-G [7]	$55.85_{0.16}$	$52.39_{0.13}$	$64.35_{0.31}$	$89.12_{0.26}$	$89.94_{0.19}$	$89.79_{0.56}$	$54.97_{0.16}$	$55.94_{0.20}$	$63.77_{0.05}$	$68.46_{0.22}$
UEO [32]	$41.18_{0.62}$	$42.39_{0.08}$	$38.63_{1.75}$	$84.70_{0.31}$	$85.51_{0.47}$	$82.73_{0.13}$	$55.58_{0.39}$	$55.55_{0.14}$	$54.26_{0.56}$	$60.06_{0.49}$
AiR [10]	$57.30_{0.85}$	$49.41_{0.27}$	$63.71_{0.21}$	$90.13_{0.31}$	$89.72_{0.50}$	$90.69_{0.20}$	$55.59_{0.27}$	$53.71_{0.47}$	$\mathbf{64.24}_{0.76}$	$68.28_{0.43}$
PPSB (Ours)	$\mathbf{59.59}_{0.70}$	$\mathbf{56.38}_{0.63}$	$\mathbf{65.51}_{1.47}$	$\mathbf{91.42}_{0.16}$	$\mathbf{91.51}_{0.11}$	$\mathbf{90.95}_{0.31}$	$\mathbf{62.85}_{0.41}$	$\mathbf{64.26}_{0.28}$	$64.11_{0.20}$	$\mathbf{71.84}_{0.47}$

F.4 Evolution of Pseudo-label Predictions

Fig. 10 shows evolution of pseudo-label predictions on the dataset Flowers102. It can be observed that our method generates pseudo-labels gradually approach balanced distribution.

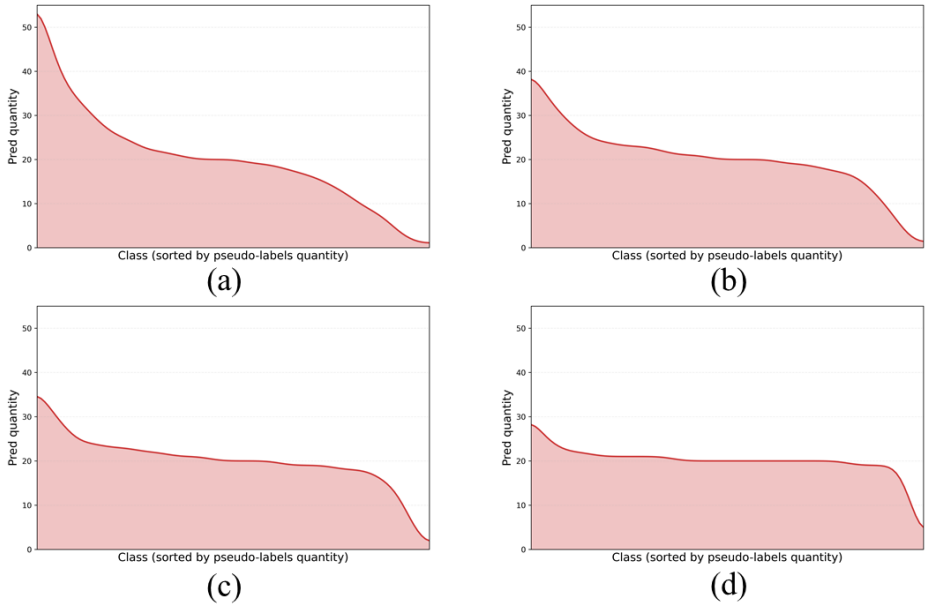


Figure 10: Evolution of Pseudo-label Predictions on the dataset Flowers102. (a)-(d) show that pseudo-label distribution approaches balanced gradually.

F.5 Evolution of Pseudo-labeled Dataset

Fig. 11 shows evolution of pseudo-label dataset on DTD. It can be observed that our method consistently construct relatively balanced pseudo-label dataset and the scale expands progressively.

F.6 Statistical Significance Test

Table 9 presents the win/tie/loss counts between our PPSB and eight baseline methods across different datasets under three settings with a pairwise t-test at a 0.05 significance level. It can be observed that

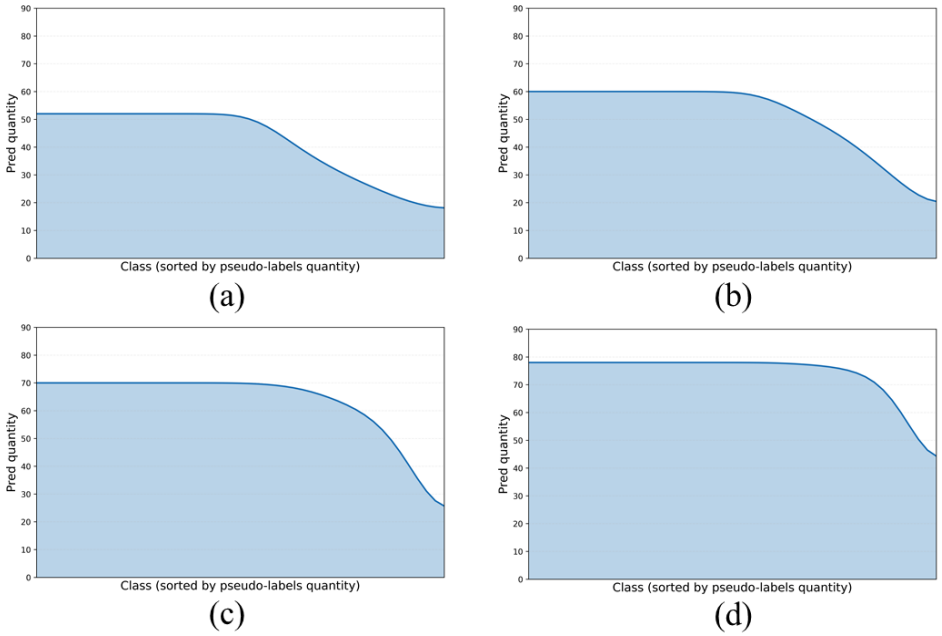
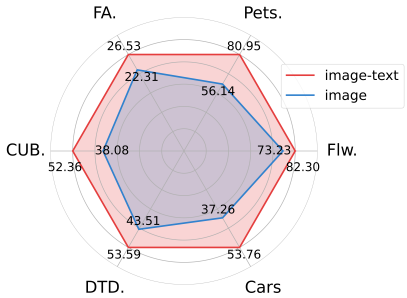


Figure 11: Evolution of Pseudo-label dataset on DTD. (a)-(d) show that pseudo-label dataset consistently remains relatively balanced and expands progressively.

Table 9: Statistical significance of performance differences assessed with pairwise t-test at a 0.05 significance level, reported as win/tie/loss counts.

Method	SSL	UL	TRZSL	Total
CLIP	6 / 0 / 0	6 / 0 / 0	6 / 0 / 0	18 / 0 / 0
FPL	6 / 0 / 0	6 / 0 / 0	6 / 0 / 0	18 / 0 / 0
GRIP	6 / 0 / 0	6 / 0 / 0	5 / 1 / 0	17 / 1 / 0
CPL-F	6 / 0 / 0	6 / 0 / 0	6 / 0 / 0	18 / 0 / 0
CPL-G	6 / 0 / 0	6 / 0 / 0	5 / 1 / 0	17 / 1 / 0
UEO	6 / 0 / 0	6 / 0 / 0	6 / 0 / 0	18 / 0 / 0
CAP	5 / 1 / 0	5 / 1 / 0	5 / 1 / 0	15 / 3 / 0
AiR	6 / 0 / 0	6 / 0 / 0	6 / 0 / 0	18 / 0 / 0
Total	47 / 1 / 0	47 / 1 / 0	45 / 3 / 0	139 / 5 / 0

Figure 12: Perform k-means on the image-text representation matrix $\mathcal{W}$ and image representation matrix $\mathcal{X}$ across six datasets. $\mathcal{W}$ consistently achieve higher clustering accuracy.



our method significantly outperforms baselines in 96.52 pp of cases (139 out of 144), demonstrating its effectiveness.

F.7 Clustering Performance of the Image-text Representation Matrix

We perform k-means on the image-text representation matrix $\mathbf{W}$ and image representation matrix $\mathbf{X}$ across six datasets. As shown in Fig. 12, $\mathbf{W}$ consistently achieves higher clustering performance than $\mathbf{X}$, which demonstrates that $\mathbf{W}$ exhibits more discriminative neighborhood information.

F.8 Comparison of Soft and Hard Pseudo-labels

Soft pseudo-labels preserve prediction uncertainty by retaining the full predicted class distribution. However, their reliability depends in part on model calibration, and non-maximum probabilities may encode erroneous correlations and propagate pseudo-label bias [7]. Hard pseudo-labels discard these probabilities and may therefore reduce the influence of unreliable class associations.

To compare the two alternatives, we replace hard pseudo-labels with soft pseudo-labels while keeping all other settings unchanged. As shown in Table 10, hard pseudo-labels achieve higher test accuracy across all six datasets, increasing the average accuracy from 58.50% to 63.37%, a gain of **4.87** percentage points. These results support the use of hard pseudo-labels as a more effective source of supervision for our method under the evaluated settings.

Table 10: Comparison of test accuracy (%) using soft and hard pseudo-labels. Results are reported as mean ± standard deviation, except for the average column. The highest accuracies are bold.

Pseudo-labels	Flw.	FA.	CUB.	DTD.	Pets.	Cars.	Avg.
Soft	76.99 ± 0.22	22.83 ± 0.30	55.20 ± 0.08	47.03 ± 0.39	88.33 ± 0.64	60.64 ± 0.08	58.50
Hard	$\mathbf{86.43} \pm 0.81$	$\mathbf{23.23} \pm 0.30$	$\mathbf{59.80} \pm 0.29$	$\mathbf{57.26} \pm 0.41$	$\mathbf{91.26} \pm 0.03$	$\mathbf{62.23} \pm 0.36$	**63.37**

F.9 Evaluation with Stronger Vision-Language Models

We use CLIP as the primary backbone to enable direct comparisons with existing unsupervised VLM adaptation methods. However, PPSB operates on image-text representations and pseudo-label distributions rather than architecture-specific components, allowing it to be applied to other stronger VLM backbones. To evaluate the generality, we conduct additional experiments using SigLIP-B/16 and PE-Core-B/16 on four datasets.

As shown in Tables 11 and 12, our method achieves the highest test accuracy on all four datasets with both backbones. On SigLIP-B/16, PPSB achieves an average accuracy of 73.81%, exceeding the strongest baseline by **1.62** percentage points. On PE-Core-B/16, its average accuracy reaches 82.82%, outperforming CAP by **4.99** percentage points. These results demonstrate the generality of PPSB across stronger vision-language models.

Table 11: Comparison results of test accuracy (%) on SigLIP-B/16.

Method	FA.	DTD.	Pets.	Cars.	Avg.
FPL [9]	37.41	65.85	92.50	88.22	71.00
GRIP [9]	27.90	65.53	92.96	82.14	67.13
CPL-F [7]	37.56	67.07	92.45	79.65	69.18
CPL-G [7]	31.44	66.17	90.92	76.14	66.17
UEO [32]	13.32	60.53	88.36	78.64	60.21
CAP [8]	39.00	68.13	93.64	87.99	72.19
AiR [10]	27.24	65.10	92.20	82.51	66.76
PPSB (Ours)	**41.55**	**70.21**	**94.23**	**89.23**	**73.81**

Table 12: Comparison results of test accuracy (%) on PE-Core-B/16.

Method	FA.	DTD.	Pets.	Cars.	Avg.
FPL [9]	56.64	66.86	94.16	92.26	77.48
GRIP [9]	49.68	68.61	93.56	89.81	75.42
CPL-F [7]	55.95	67.23	94.14	90.16	76.87
CPL-G [7]	51.78	68.35	93.81	89.31	75.81
UEO [32]	46.86	68.24	94.19	84.65	73.49
CAP [8]	53.23	72.12	94.76	91.22	77.83
AiR [10]	48.54	69.41	93.29	90.33	75.39
PPSB (Ours)	**68.34**	**73.77**	**95.33**	**93.84**	**82.82**

G Limitations

Although the proposed progressive pseudo-label self-balancing framework effectively avoids numerous incorrect pseudo-tail labels, further improving the accuracy of pseudo-tail classes remains a challenging task. In the future work, we aim to refine our framework to better address this issue.

H Compute Resources

- CPU: AMD EPYC 7642 48-Core Processor × 2
- GPU: NVIDIA GeForce RTX 4090 24G × 1
- MEM: 500G
- Maximum total computing time per dataset: training + testing ≈ 30min.

I Broader Impacts

This paper presents work whose goal is to advance the field of Machine Learning. Specifically, we divide all classes into pseudo-head, pseudo-middle, and pseudo-tail classes based on pseudo-label

quantity. Accordingly, we propose a progressive pseudo-label self-balancing framework, which measures the class-wise prediction bias and calculates quantity upper bounds for certain classes to construct relatively balanced pseudo-labeled dataset. Furthermore, we design an image-text neighborhood filtering mechanism and a visual classifier to generate high-quality pseudo-labels. We also theoretically prove that our method can significantly reduce the generalization error under certain conditions. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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